

MindSpine

Business Requirements Document (BRD) — Physiotherapy Workflow

Appointment Booking • Patient History • Physiotherapy Care • Payments • Reports

Purpose: Define the business and functional requirements for the MindSpine physiotherapy module, covering patient access, physiotherapist workflow, appointment management, clinical documentation, billing, notifications, reporting, Excel/PDF export and administration.

1. Business Overview

MindSpine will provide a role-based digital workflow for physiotherapy practices. Patients can register, book and pay for appointments, receive reminders, view shared reports and bills, and track care. Physiotherapists can securely access patient history, document assessments and treatment notes, create rehabilitation plans, manage appointments, generate reports and export permitted data. Administrators manage users, schedules, configuration and operational oversight.

2. Business Goals

- Digitize the physiotherapy journey from booking through follow-up.
- Give physiotherapists a longitudinal view of patient history and treatment progress.
- Reduce manual scheduling, billing, reminders and report preparation.
- Support online booking and payment through an integrated payment gateway.
- Provide printable patient bills, invoices and physiotherapy assessment/progress reports.
- Allow authorized PDF and Excel exports.
- Maintain secure, role-based access to patient and clinical information.
- Create a scalable base for future rehabilitation and analytics features.

3. User Roles

Role	Requirements
Patient / User	Register/login; maintain profile; book/reschedule/cancel; pay online; receive notifications; view shared care information, bills and reports.
Physiotherapist	View patient history; assess; document findings and treatment notes; create rehabilitation plan; update progress; manage appointments; generate reports; export authorized data.
Admin	Manage users and physiotherapist profiles; schedules; appointment configuration; payment/report settings; analytics; access controls and audit oversight.

4. Core Physiotherapy Workflow

Step	Activity	Description
1	Login & access	Secure patient/physio/admin access with role permissions.
2	Patient profile	Demographics, consent and relevant history captured.
3	Appointment booking	Patient selects service, physiotherapist and date/time.
4	Payment	Online payment processed; transaction status and receipt recorded.
5	Pre-session history	Physiotherapist reviews history, prior visits and documents.
6	Assessment	Complaint, pain/function measures, findings and clinical impression recorded.
7	Rehabilitation plan	Goals, interventions, exercises, frequency and milestones documented.

Step	Workflow	Outcome
8	Session documentation	Treatment, response, progress and next actions captured.
9	Billing & report	Bill/invoice and physiotherapy report generated for print/PDF.
10	Follow-up	Reminders sent; progress tracked and plan adjusted.

5. Functional Requirements

5.1 Patient / User Features

ID	Requirement
P01	Sign up/login using email or phone; password reset and optional MFA/2FA.
P02	Create/update patient profile and consent.
P03	Book online or in-clinic physiotherapy appointments.
P04	Choose available slot and preferred physiotherapist where enabled.
P05	Reschedule/cancel according to clinic rules.
P06	Make online payment via configured payment gateway.
P07	View payment status and receive invoice/receipt.
P08	Receive booking, reminder, reschedule and cancellation notifications.
P09	View shared rehabilitation plan and session documents.
P10	View/download permitted bills and reports in PDF.
P11	View own visit history/timeline as permitted.

5.2 Physiotherapist Features

ID	Requirement
PH01	Secure login and professional profile with credentials, specialization, availability and fee.
PH02	View assigned/upcoming appointments and manage availability.
PH03	Search patients and view complete authorized patient history.
PH04	Review previous sessions, reports, uploads and rehabilitation plans.
PH05	Record complaint, history, pain/function measures, observations, examination findings and clinical impression.
PH06	Create personalized rehabilitation plan with goals, exercises, frequency and milestones.
PH07	Record structured treatment/session notes and patient response.
PH08	Update progress at each visit and modify plan when needed.
PH09	Upload supporting documents/images/reports.
PH10	Generate assessment, progress and discharge reports.
PH11	Generate/associate bill or invoice with appointment.
PH12	Print/download reports and bills; export authorized data to Excel.
PH13	Schedule next visit and trigger follow-up/reminder notifications.

5.3 Admin Features

ID	Requirement
A01	Create, activate, deactivate and manage accounts.
A02	Verify and maintain physiotherapist credential information.
A03	Configure services, appointment duration, working hours, holidays and scheduling rules.
A04	Configure gateway, fees, cancellation/refund rules and invoice settings.
A05	Manage notification templates and channels.
A06	View operational analytics for appointments, payments and reports.
A07	Manage roles, permissions, audit logs and platform configuration.

ID	Requirement
A08	Configure report templates, export permissions and retention rules.

6. Clinical Data & Documentation

Item	Details / Capabilities
Patient history	Demographics, relevant medical history, injuries/conditions, prior visits, documents and treatment history.
Assessment	Presenting complaint, onset, pain score, functional limitations, observation, range-of-motion/strength or configured measures, red-flag notes and clinical impression.
Treatment plan	Goals, interventions, exercises, frequency, duration, precautions and review date.
Session note	Treatment delivered, exercise progression, response, pain/function changes and next plan.
Progress tracking	Outcome-measure trends, milestone status, plan changes and progress narrative.
Reports	Assessment, progress, treatment summary/discharge, bill/invoice and receipt.

7. Appointment, Payment & Notification Requirements

- **Appointments:** real-time slot availability, confirmation, reschedule/cancel and status tracking.
- **Payments:** initiation, success/failure/pending status, webhook confirmation, transaction reference, invoice/receipt and configured refunds.
- **Notifications:** booking confirmation, appointment reminder, payment status, reschedule/cancellation, report-ready and follow-up alerts.
- **Channels:** email, SMS and/or push; WhatsApp can be enabled through an approved provider.
- **Print:** printable bill/invoice and printable physiotherapy assessment/progress report.

8. Reporting & Export

Report	Format / Features
Patient bill / invoice	Printable/downloadable PDF with patient, service, appointment, amount, taxes/discounts if applicable, payment status and reference.
Physiotherapy report	Printable/downloadable PDF with assessment, clinical impression, treatment plan, progress and therapist details.
Patient history export	Authorized Excel export of selected patient/visit data subject to role and consent.
Appointment report	Excel/PDF export by date range, status, service and physiotherapist.
Payment report	Excel/PDF export of transactions, status and reconciliation fields.
Audit report	Administrative access to critical activity logs subject to authorization.

9. Integrations

Integration	Purpose
Payment gateway	Razorpay / Stripe / PayU or selected provider for payments, webhooks and refunds.
Email service	Transactional confirmations, reminders and report-ready notifications.
SMS / WhatsApp	Appointment and payment notifications where enabled.
Calendar	Optional Google Calendar / Outlook synchronization.
File storage	Secure storage for reports, images and patient documents.
Monitoring	Operational analytics, system monitoring and error tracking.

10. Non-Functional Requirements

	Requirements
NFR01	Security: HTTPS/TLS, encryption at rest where applicable, secure credentials and least-privilege access.
NFR02	Privacy: consent, role-based visibility, controlled exports and retention/deletion policies.
NFR03	Auditability: log critical access and changes to clinical, payment and admin records.
NFR04	Availability: production architecture should support resilient deployment; target SLA finalized in solution design.
NFR05	Performance: common screens/APIs optimized for responsive clinic use; thresholds finalized in solution design.
NFR06	Scalability: support growth in patients, appointments, files and concurrent users.
NFR07	Usability: responsive web/mobile experience with low-friction clinical workflows.
NFR08	Accessibility: follow applicable accessibility standards.
NFR09	Backup & recovery: automated backups and tested restore process.
NFR10	Compliance: review data handling against applicable healthcare/privacy obligations in operating jurisdictions.

11. High-Level Data Entities

- User, Role, Patient Profile, Physiotherapist Profile
- Appointment, Availability Slot, Service
- Patient History, Assessment, Treatment Plan, Session Note, Progress Measure
- Document / Attachment, Clinical Report
- Invoice / Bill, Payment Transaction, Refund
- Notification, Consent, Audit Log

12. Business Rules

- Only authorized roles may access clinical information; patients access only their own permitted records.
- Physiotherapist access follows assignment/clinic access rules.
- Clinical notes retain author and timestamp; material edits are auditable.
- Appointment slots cannot be double-booked unless explicitly configured.
- Payment success must be confirmed by the provider before an appointment is treated as paid.
- Exports honor role permissions, consent, selected scope and data-minimization rules.
- Report and invoice numbers are unique within the configured business context.
- Cancellation, refund and no-show rules are clinic-configurable.

13. Acceptance Criteria

	Acceptance Criteria
1	Patient, physiotherapist and admin can securely log in with correct role permissions.
2	Patient can book an available physiotherapy slot and receive confirmation.
3	Online payment can be completed and transaction status stored.
4	Physiotherapist can view authorized patient history before a session.
5	Physiotherapist can save assessment, treatment plan and session notes.
6	Progress can be updated across multiple visits.
7	Bill/invoice and physiotherapy report can be generated and printed/downloaded.

	Requirements
8	Authorized report/patient data can be exported to Excel.
9	Patient receives configured appointment and payment notifications.
10	Admin can manage users, schedules, configuration and permissions.
11	Critical clinical/payment/admin actions are auditable.
12	Unauthorized users cannot access protected patient or clinical data.

14. Future Enhancements

- Patient exercise library with video instructions and adherence tracking.
- Body-map / pain-map visualization and longitudinal outcome dashboards.
- AI-assisted session-note summarization with clinician review.
- Native mobile application and wearable integration for activity/recovery data.
- Tele-rehabilitation/video consultation integration.
- Patient self-care recommendations and home-exercise reminders.
- Insurance/EHR integrations where commercially and legally appropriate.
- Multi-language support and advanced clinic analytics.

15. Out of Scope for Initial Release

- Automated clinical diagnosis without physiotherapist review.
- Emergency medical decision-making.
- Direct replacement of regulated clinical judgment.
- Insurance claim adjudication unless separately integrated and specified.
- Advanced AI recommendations until governance, validation and human-review requirements are defined.

16. Physiotherapy workflow revision

Revision 1.1 | 6 October 2026. This addendum supplements the original BRD on pages 1-7. The original requirements are retained. Where evaluation-form requirements differ, this revision takes precedence: keep both existing forms and do not replace or redesign their fields.

16.1 Two evaluation options - unchanged

The physiotherapist selects either **Standard physiotherapy** or **Chiropractic evaluation**. Both remain available in the same evaluation selector. The existing evaluation fields, clinical summary, treatment plan, patient-release control and progress sheet remain unchanged. Psychology workflows are separate.

16.2 Care sequence

Step	Required behavior
1. Patient and history	Add/select an authorized patient. Review demographics, medical history, prior visits, evaluations and supporting documents.
2. Appointment	Choose an available clinician slot. Booking creates the associated invoice. Clinic staff record offline payment; payment status is not inferred from booking.
3. Evaluation	Complete either existing evaluation form for the selected visit. Save a draft or release to the patient.
4. Rehabilitation plan	Link a dated plan to the evaluation. Record goals, interventions, home exercises, frequency/duration, precautions, milestones and review date.
5. Session progress	Append dated treatment delivered, patient response, pain score (optional, 0-10), functional progress, milestones and next actions.
6. Discharge and follow-up	Append a discharge summary and advice. Record review date, book the next visit separately and download or print reports.

17. Chiropractic evaluation requirements

The chiropractic evaluation follows the clinic reference supplied by the user. It is an alternative evaluation within the physiotherapist workflow, not a new user role. No field layout or field meaning is changed by this workflow revision.

Section	Existing fields retained
Patient details	Evaluation date, age, gender, contact, address and profession; patient linked through the selected visit.
History	History of present illness (HPI), medical history, surgical history and family history.
Examination	Blood pressure (BP), heart rate (HR), weight and height.
Observation	Scar (if any), posture, gait, deformity and edema.
Palpation	Tenderness, scar mobility, tone, spasm and trigger points.
Manual muscle testing	SIC, WIC, TL, facilitated and inhibited. Labels are preserved as supplied; no expansion is inferred.
Assessment and treatment	Special tests, range of motion (ROM), chief complaint (C/C), diagnosis or clinical summary and treatment plan.
Progress sheet	Numbered entries with date and treatment/progress narrative.
Report release	Explicit clinician choice to release the report to the patient. Drafts remain hidden from patients.

17.1 Reports and exports

A chiropractic report must identify itself as Chiropractic evaluation and include the recorded evaluation fields and progress sheet. Standard evaluations retain their own report title. Rehabilitation plans, session-progress reports and discharge summaries are separate printable PDFs linked to their source evaluation. Authorized Excel exports include both evaluation details and rehabilitation records.

18. Acceptance and delivery boundaries

18.1 Acceptance checks for this revision

- Both evaluation options remain available with their original fields unchanged.
- A physiotherapist can add a patient, review their history, book a visit and choose either evaluation.
- A saved evaluation can have multiple dated rehabilitation plans, session-progress records and discharge summaries. Prior records remain unchanged.
- Only a physiotherapist with access to the patient can create rehabilitation records. Administrator operational access does not expose clinical narratives.
- Patients see only their own released evaluations and rehabilitation records. A draft evaluation cannot be used to release a rehabilitation record.
- Each workflow save records the author, timestamp and audit event. Review dates create in-app clinician notices; booking is a separate confirmation step.
- Physiotherapists can view/download invoices only for their own appointments. Payment/refund recording remains administrator-only.
- PDF and Excel exports preserve the evaluation type and linked rehabilitation details. Unauthorized downloads are rejected.

18.2 Integration and broader BRD requirements

The source BRD remains the target specification. This revision does not claim that every target integration is delivered. The current implementation uses email/password authentication, in-app notifications and clinic-recorded offline payments.

Pending provider setup: online gateway checkout/webhooks/refunds, external email/SMS/WhatsApp delivery and calendar synchronization. These require selected providers and credentials; no simulated transaction is marked as provider-confirmed payment.

Remaining broader requirements: phone login/MFA, formal consent and retention management, credential verification, configurable service fees/durations/holiday rules, automated timed reminders, appointment/payment reports with date filters, backup/restore validation and production compliance review. These remain requirements, not completed acceptance claims.

Form preservation: rehabilitation workflow fields are outside the evaluation form. Standard physiotherapy and chiropractic evaluation remain the source clinical evaluations without redesign.